RAJALAKSHMI ENGINEERING COLLEGE

Kanishk P 251801127-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 10:31 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:35 AM
Time taken	4 mins 24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int count[100001] = {0};
    int x;

    for (int i = 0; i < n; i++) {
        scanf("%d", &x);

        count[x]++;

        if (count[x] == 2) {
            printf("%d", x);
            return 0;
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Kanishk P 251801127-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 10:36 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:40 AM
Time taken	4 mins 40 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>

int main()
{
    int n;
    scanf("%d",&n);

    int arr[n];

    int count[1001]={0};

    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);

        count[arr[i]]++;

        if(count[arr[i]]==2)
        {
            printf("%d",arr[i]);
            return 0;
        }
    }
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Kanishk P 251801127-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 10:41 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:05 AM
Time taken	24 mins 39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int T;
    scanf("%d", &T);

    while (T--)
    {
        int n, m;

        scanf("%d", &n);

        int arr1[n];
```

```
15
16          for (int i = 0; i < n; i++)
17          {
18              scanf("%d", &arr1[i]);
19          }
20
21          scanf("%d", &m);
22
23          int arr2[m];
24
25          for (int i = 0; i < m; i++)
26          {
27              scanf("%d", &arr2[i]);
28          }
29
30          int i = 0;
31          int j = 0;
32          int first = 1;
33
34          while (i < n && j < m)
35          {
36              if (arr1[i] == arr2[j])
37              {
38                  if (!first)
39                      printf(" ");
40
41                  printf("%d", arr1[i]);
42
43                  first = 0;
44                  i++;
45                  j++;
46              }
47              else if (arr1[i] < arr2[j])
48              {
49                  i++;
50              }
51              else
52              {
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Kanishk P 251801127-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 11:05 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:17 AM
Time taken	11 mins 45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>

int main()
{
    int T;
    scanf("%d",&T);

    while(T--)
    {
        int n;
        scanf("%d",&n);
        int arr1[n];

        for(int i=0;i<n;i++)
```

```
        {
            scanf("%d",&arr1[i]);
        }

        int m;
        scanf("%d",&m);
        int arr2[m];

        for(int j=0;j<m;j++)
        {
            scanf("%d",&arr2[j]);
        }

        int i = 0;
        int j = 0;
        int first = 1;

        while (i < n && j < m)
        {
            if (arr1[i] == arr2[j])
            {
                if (!first)
                    printf(" ");

                printf("%d", arr1[i]);

                first = 0;
                i++;
                j++;
            }
            else if (arr1[i] < arr2[j])
            {
                i++;
            }
            else
            {
                j++;
            }
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Kanishk P 251801127-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 11:18 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:34 AM
Time taken	15 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0;
    int j = 1;

    while (i < n && j < n)
    {
        if (i == j)
        {
            j++;
        }
        else if (arr[j] - arr[i] == k)
        {
            printf("1");
            return 0;
        }
        else if (arr[j] - arr[i] < k)
```

```
        {
            j++;
        }
        else
        {
            i++;
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Kanishk P 251801127-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 11:25 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:26 AM
Time taken	59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0;
    int j = 1;

    while (i < n && j < n)
    {
        if (i == j)
        {
            j++;
        }
        else if (arr[j] - arr[i] == k)
        {
            printf("1");
            return 0;
        }
        else if (arr[j] - arr[i] < k)
```

```
        {
            j++;
        }
        else
        {
            i++;
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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